AMENDMENT

TO

INTERIM FEDERAL SPECIFICATION

WRENCH, OPEN END BOX (AND BOX COMBINATION CROWFOOT, RATCHET, AND RATCHET CROWFOOT)

This amendment was developed by the Standardization Division, Federal Supply Service, General Services Administration, Washington, D. C. 20405, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this amendment as a valid exception to Interim Federal Specification GGG-W-00645b, dated October 30, 1967.

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Paragraph 4.6.2.2 - Delete and substitute:

4.6.2.2 Tube fitting distortion test. The open-end of types I, II, III, IV and V wrenches shall conform to table IV when subjected to the following tube fitting distortion test. Place a single or double hex socket wrench of the same size of the open-end wrench to be tested into a vise or torque analyzer. Insert into the socket the corresponding size threaded steel coupling conforming to MIL-F-18866B. A NF-3, GO and NO-GO thread gauge shall be used, making sure that the NO-GO end of the gauge does not enter into the steel coupling until it bottoms. Be sure the fitting is free of imperfections, and the GO gauge turns freely the full length of the fitting. A light oil should be used on the threads for this test to minimize thread variables. After threading the gauge to the bottom of the coupling, back the gauge up one full turn. Place the wrench on the fitting with the center of the wrench as near to the center of the fitting as possible (see figure 6) by inserting the fitting only as far into the socket as is neccesary to hold it securely, (approximately 1/8 inch) when the torque is applied to it. The wrench shall then be brought up to approximately one half the torque, for that size in table IV.. This shall be done three times at one half torque, rotating the wrench 30° to the next gripping position of the wrench, after each single torque cycle. (In the case of the ratchet to the next ratchet position.) Then while constantly rotating the thread gauge, clockwise and counterclockwise, by hand, through a short arc (1/16 to 1/8 turn), sufficient to merely keep the gauge in motion, bring the wrench up to the torque, where the gauge can no longer be forced to turn by hand. Repeat this procedure a total of six times, rotating the wrench 30° to the next gripping position of the wrench, after each single torque cycle, as before. (In case of the ratchet, to the next ratchet position.) The three lowest torque readings, at which the thread gauge could no longer be turned by hand, should be averaged. This torque figure for each shall be no lower than the figure shown in table IV.

TABLE IV. Torque distortion test

Wrench size	Torque
Inch	Foot - Pounds
3/8	20
7/16	21
1/2	25
9/16	28
5/8	30
11/16	45
3/4	48
7/8	50
15/16	55
1	60
1-1/8	65

NOTE: O. D. Limits of fitting for this test, shall be no more than ±0.001 inch.

GSA DC 69-4748

FSC 5120

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